

PathFinder — Sprint Logs

Sprint 1: Foundation (Week 1)

Sprint Goal

Establish the project foundation including project setup, database design, authentication, and basic CRUD operations.

Planned Tasks

- [x] Initialize frontend with Vite + React + TypeScript
- [x] Initialize backend with Express.js + TypeScript
- [x] Configure Prisma ORM with SQLite
- [x] Design complete database schema (16 tables)
- [x] Implement user registration and login
- [x] Implement JWT authentication middleware
- [x] Implement role-based authorization middleware
- [x] Create basic student profile CRUD
- [x] Create basic organization profile CRUD
- [x] Set up file upload system with Multer
- [x] Implement password reset flow

Completed Tasks

All planned tasks completed.

Challenges

- SQLite file encoding issues when creating files via PowerShell (UTF-16 LE BOM). Resolved by rewriting affected files with proper encoding.

Solutions

- Rewrote the stats.ts route file with correct UTF-8 encoding to resolve the backend crash.

Sprint 2: Core Features (Week 2)

Sprint Goal

Build the primary user-facing features: dashboards, applications, search, notifications.

Planned Tasks

- [x] Create Landing page with premium design
- [x] Build Student Dashboard with statistics
- [x] Build Organization Dashboard
- [x] Implement opportunity CRUD operations
- [x] Build application submission flow
- [x] Implement application status management
- [x] Create application timeline system
- [x] Build search and filter page
- [x] Implement bookmarking system
- [x] Create notification system
- [x] Design complete CSS design system (glassmorphism, dark mode)
- [x] Build DashboardLayout with sidebar navigation
- [x] Implement ProtectedRoute component

Completed Tasks

All planned tasks completed.

Challenges

- Managing the complexity of role-based sidebar navigation with different menu items for each role.
- Ensuring the glassmorphic design system looked premium on both dark and light modes.

Solutions

- Centralized menu item configuration with role filtering in DashboardLayout.
 - Used CSS custom properties with data-theme attribute for seamless theme switching.
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Sprint 3: AI & Advanced Features (Week 3)

Sprint Goal

Integrate AI features, build the admin dashboard, and implement the support ticketing system.

Planned Tasks

- [x] Implement AI match scoring engine
- [x] Build resume PDF parsing
- [x] Create SOP generator
- [x] Create Cover Letter generator
- [x] Create Essay generator
- [x] Create Personal Statement generator
- [x] Build profile strength analysis
- [x] Implement opportunity recommendations
- [x] Build AI candidate summary for orgs
- [x] Create comprehensive Admin Dashboard
- [x] Implement user management (search, filter, roles, delete)
- [x] Build audit log system
- [x] Create system health monitor
- [x] Implement CSV data export
- [x] Build support ticketing system
- [x] Implement threaded ticket messaging

Completed Tasks

All planned tasks completed.

Challenges

- Designing the AI system to work both with and without external API keys.
- Ensuring asynchronous match scoring doesn't block the application submission response.

Solutions

- Built a dual-mode architecture with intelligent mock fallbacks.
 - Implemented fire-and-forget async pattern for background match scoring.
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Sprint 4: Deployment & Documentation (Week 4)

Sprint Goal

Deploy to production and create comprehensive documentation.

Planned Tasks

- [x] Deploy backend to Render
- [x] Deploy frontend to Vercel
- [x] Configure environment variables on both platforms
- [x] Override Vercel build command to skip strict TypeScript checks
- [x] Test production deployment end-to-end
- [x] Create 20-document documentation suite
- [x] Generate PDF versions of all documents
- [x] Push all documentation to GitHub

Completed Tasks

All planned tasks completed.

Challenges

- Vercel build failure due to TypeScript strict mode catching unused imports.

- Render free tier cold starts causing initial request delays.

Solutions

- Overrode Vercel build command from `tsc -b && vite build` to `vite build`.
- Documented the cold start behavior in user-facing materials.

End of Sprint Logs